

Project_1_Data_Cleaning

June 10, 2026

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[3]: import pandas as pd
import os

raw_data = "Dataset for Data Analytics.xlsx"

if os.path.exists(raw_data):
    # Use read_excel to cleanly parse the spreadsheet layers
    df = pd.read_excel(raw_data)
    print(" Excel environment successfully established!")
    print(f" Dataset Shape: {df.shape[0]} transaction rows loaded.")
else:
    print(f" Error: Could not find '{raw_data}' in your folder workspace.")
    print("Double check that the file name is spelled exactly right (including_
↳capitals).")
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Excel environment successfully established!
Dataset Shape: 1200 transaction rows loaded.
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[ ]: # Print out a summary of missing values across all columns
print("--- Missing Values Audit ---")
print(df.isnull().sum())

# Check for duplicate Order IDs to test our unique identifier gate status
print("\n--- Unique ID Verification Check ---")
print(f"Total Rows in Dataset: {len(df)}")
print(f"Unique OrderID Counts: {df['OrderID'].nunique()}")
print(f"Rogue Duplicate OrderID Rows: {df['OrderID'].duplicated().sum()}")
```

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--- Missing Values Audit ---
OrderID          0
Date             0
CustomerID       0
Product          0
Quantity         0
UnitPrice        0
ShippingAddress  0
PaymentMethod    0
OrderStatus      0
TrackingNumber   0
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ItemsInCart      0
CouponCode       309
ReferralSource   0
TotalPrice       0
dtype: int64

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--- Unique ID Verification Check ---
Total Rows in Dataset: 1200
Unique OrderID Counts: 1200
Rogue Duplicate OrderID Rows: 0

```

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[5]: # Defensive Text Layer Scrubbing (Clean Leading/Trailing Spaces)
string_columns = df.select_dtypes(include=['object']).columns
for col in string_columns:
    df[col] = df[col].astype(str).str.strip()

[6]: # Structural Imputation of Missing Column Blocks
df['CouponCode'] = df['CouponCode'].replace(['nan', 'NaN', 'None'], np.nan)
df['CouponCode'] = df['CouponCode'].fillna('NO_COUPON')

[7]: # Mandatory Gate Check: Force Strict ISO 8601 Temporal Format (0% Format Error)
df['Date'] = pd.to_datetime(df['Date'], errors='raise')
df['Date'] = df['Date'].dt.strftime('%Y-%m-%d')

[8]: # Mandatory Gate Check: Enforce Absolute Deduplication (0% Duplicate ID Error)
df = df.drop_duplicates(subset=['OrderID'], keep='first')

[9]: # Numeric Precision Engineering (Force Uniform 2-Decimal Precision)
df['UnitPrice'] = df['UnitPrice'].round(2)
df['TotalPrice'] = df['TotalPrice'].round(2)

[10]: # Final Sanity Check Assertions
assert df['OrderID'].duplicated().sum() == 0, " Error Gate Violated: Duplicate_
↳unique IDs found!"
assert pd.to_datetime(df['Date'], errors='coerce').notnull().all(), " Error_
↳Gate Violated: Bad date formatting!"

[13]: print("=====")
print("          PRE-EXPORT DATA INTEGRITY REPORT          ")
print("=====")

# 1. STRUCTURAL DIMENSION AUDIT
print(f" Dataset Dimensions: {df.shape[0]} rows | {df.shape[1]} columns")
if df.shape[0] < 1000:
    print(" WARNING: Row count dropped significantly! Double-check your_
↳filters.")
else:
    print(" Row Count: Stable and safe.")

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# 2. MANDATORY GATE 1: UNIQUE IDENTIFIER CHECK (0% Error Rate)
duplicate_ids = df['OrderID'].duplicated().sum()
print(f"\n Duplicate OrderIDs found: {duplicate_ids}")
if duplicate_ids == 0:
    print(" GATE 1 PASSED: Unique identifier error rate is exactly 0%.")
else:
    print(" GATE 1 FAILED: Duplicates still exist in primary keys!")

# 3. MANDATORY GATE 2: DATE COMPLIANCE CHECK (0% Error Rate)
# Coerce dates to see if any failed to format (they will turn into NaT / Null)
null_dates = pd.to_datetime(df['Date'], errors='coerce').isnull().sum()
print(f"\n Malformed Date entries found: {null_dates}")
if null_dates == 0:
    print(" GATE 2 PASSED: All dates are perfectly uniform (ISO 8601).")
else:
    print(" GATE 2 FAILED: You have malformed date strings remaining!")

# 4. MISSING VALUES POST-AUDIT
print("\n Remaining Null/Missing Values Per Column:")
print(df.isnull().sum())

# 5. DATA TYPES & FORMAT PREVIEW
print("\n Data Type Blueprint:")
print(df.dtypes)

print("\nVisual Verification (First 3 Records):")
display(df.head(3))

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=====
PRE-EXPORT DATA INTEGRITY REPORT
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Dataset Dimensions: 1200 rows | 14 columns
Row Count: Stable and safe.

Duplicate OrderIDs found: 0
GATE 1 PASSED: Unique identifier error rate is exactly 0%.

Malformed Date entries found: 0
GATE 2 PASSED: All dates are perfectly uniform (ISO 8601).

Remaining Null/Missing Values Per Column:

OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0

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ShippingAddress    0
PaymentMethod      0
OrderStatus        0
TrackingNumber     0
ItemsInCart        0
CouponCode         0
ReferralSource     0
TotalPrice         0
dtype: int64
```

Data Type Blueprint:

```
OrderID            object
Date              object
CustomerID         object
Product            object
Quantity           int64
UnitPrice          float64
ShippingAddress    object
PaymentMethod      object
OrderStatus        object
TrackingNumber     object
ItemsInCart        int64
CouponCode         object
ReferralSource     object
TotalPrice         float64
dtype: object
```

Visual Verification (First 3 Records):

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	\
0	928 Main St	Debit Card	Shipped	TRK37947903	7	
1	823 Main St	Online	Shipped	TRK91186779	3	
2	512 Main St	Credit Card	Cancelled	TRK42903982	8	

	CouponCode	ReferralSource	TotalPrice
0	SAVE10	Instagram	2853.1
1	SAVE10	Referral	302.7
2	FREESHIP	Email	2753.4

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[14]: # Export the High-Integrity Golden Record
clean_output_path = "Clean_Dataset_Data_Analytics.csv"
df.to_csv(clean_output_path, index=False)
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print("\n --- VERIFICATION GATE QUALITY REPORT ---")
print(f" Total Verified Rows Saved: {len(df)}")
print(f" Unique ID Error Rate: {df['OrderID'].duplicated().sum()}%")
print(f" Missing Date Format Errors: {df['Date'].isnull().sum()}")
print(f" Clean File Output Generated Successfully: {clean_output_path}")
```

--- VERIFICATION GATE QUALITY REPORT ---

Total Verified Rows Saved: 1200

Unique ID Error Rate: 0%

Missing Date Format Errors: 0

Clean File Output Generated Successfully: Clean_Dataset_Data_Analytics.csv